Corrected and conduct of chapter 5

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $\frac{12}{40} = \frac{3}{10} = 30\,\%$. *Action:* Partial and reference effective linkage.
2. $0{,}7$. *Action:* Represent the universe shared in two events.
3. $A \cap B$. *Action:* Coloring the intersection of two sets.
4. $A \cup B$. *Action:* Include common issues in the meeting.
5. $0{,}1$. *Action:* Replace the decimals with fractions.
6. $\frac{3}{6} = \frac{1}{2}$. *Action:* Identify the equiprobability hypothesis.

Research activity

Intersection required 12; boxes 12,28,18,42. Replace by 13,27,17,43 keeps margins but $\frac{13}{100} \neq 0{,}4 \times 0{,}3$.

Immediate applications

1. $\frac{5}{60} = \frac{1}{12}$ and $\frac{15}{20} = \frac{1}{4}$.
2. $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.
3. No: 0 $,4 \times 0{,}5 = 0{,}20 \neq 0{,}18$.
4. $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.
5. $\frac{675}{2000} = 0{,}3375$; this is an estimate, not the exact value.

Exercises: approach and results

Exercise 5.1 · A table. Boxes: 15 (A and B), 25 (A without B), 20 (B without A), 40 (Neither A nor B).
$P_A(B) = \frac{15}{40} = \frac{3}{8}$; $P_B(A) = \frac{15}{35} = \frac{3}{7}$.

Exercise 5.2 · Production. $P(D) = 0{,}25 \times 0{,}04 + 0{,}75 \times 0{,}02 = 0{,}025$. $P_D(A) = \frac{0{,}01}{0{,}025} = 0{,}4$.
The denominator changes when it is conditioned by D.

Exercise 5.3 · Independence to be checked. $P(A) = P(B) = \frac{1}{2}$; intersection $\{4; 6\}$ probability $\frac{1}{-} \neq \frac{1}{-}$: The fact that another threshold choice gave independence does not transfer
Not automatically.

Exercise 5.4 · No remission. Two reds: ($\frac{3}{5})(\frac{2}{4}) = \frac{3}{10}$. One of each: ($\frac{3}{5})(\frac{2}{4}) + (\frac{2}{5})(\frac{3}{4}) = \frac{3}{5}$. The order distinguishes two disjointed paths.

Exercise 5.5 · Three tests. One success: three paths, each ($\frac{1}{3})(\frac{2}{3})^2 = \frac{4}{27}$Total $\frac{4}{9}$. Two success: $3(\frac{1}{3})^2(\frac{2}{3}) = \frac{2}{9}$; three : $\frac{1}{27}$; Total $\frac{7}{27}$.

Exercise 5.6 · Detector. $P(S) = 0{,}02(0{,}9)+0{,}98(0{,}05) = 0{,}067$. $P_S(D) = \frac{0{,}018}{0{,}067} = \frac{18}{67} \approx 26{,}9\,\%$. The initial low rate explains the difference with 90%.

Exercise 5.7 · Together and Cardinal. $A \cap B = \{2\}$ (1) ; $A \cup B = \{1;2;4\}$ (3) ; $E \setminus A = \{3;4\}$ (2); product $\{(1;2),(1;4),(2;2),(2;4)\}$ (4) A couple is ordered.

Exercise 5.8 · Deferred review. $P(B) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$. $P_B(A) = \frac{\frac{1}{8}}{\frac{1}{2}} = \frac{1}{4}$. Digital equality with $P_A(B)$ is particular to these data.

CQFD, Exercise 67 p. 336, Questions 1, 2 and 3. $P(\overline{D} \cap \overline{N}) = 0{,}9 \times 0{,}05 = 0{,}045$. Probability a positive test is 0 ,$1 \times 0{,}97 + 0{,}045 = 0{,}142$. Among the positive tests, the probability of not be doped is worth $\frac{0{,}045}{0{,}142} = \frac{45}{142} \approx 31{,}7\%$ Do not reverse positive and negative test.

CQFD, exercise 70 p. 336, questions a and b. $P(B) = \frac{1}{4}$, $P_A(B) = \frac{1}{2}$ and $P_{\overline{A}}(B)$ = 0. $\frac{1}{4} \neq \frac{1}{2}$ + 0, then 0 $\neq 1 - \frac{1}{2}$. The formula of total probabilities includes weights $P(A)$ and $P(\overline{A})$; the supplement applies to B when the same condition is laid down.

MCQ: answers and reasons

MCQ 1.

1. B. Reference universe A.

2. A. Product of branches.

3. D. Definition.

4. C. Criterion without division.

5. B. Zero intersection, positive product.

MCQ 2.

1. C. Supplement of three sides.

2. A. Output p^2.

3. D. Probabilities of conditional partition.

4. B. Observed frequency and probability of the model differ.

5. C. There are 2 reds on 4 balls.

True or false

1. False: 12 common objects out of 40 A and 30 B give $\frac{3}{10}$ and $\frac{2}{5}$.

2. True: $P(A \cap \overline{B}) = P(A) - P(A \cap B) = P(A)(1 - P(B))$.

3. False: without remission, the composition of the urn changes.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.